SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Dashboard Development Task

Course Code:	DSA0502	**Course Title:**	Query Processing for Data Science
CO Assessed:	CO5	**Assessment:**	Tool 3– Dashboard Development Task
Weightage:	25%	**Total Marks:**	
CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication			
Student Name:	B.Varun Kumar Reddy	**Reg. No.:**	192324066
Section / Batch:	AIDS 4th year	**Date of Submission:**	21/08/26
Faculty In-charge:	**Dr. B.Shyamala Gowri**	**Project Mode:**	Individual Submission

Code of Conduct

I B.Varun Kumar Reddy(192324066) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: ___________________________

Question 1 – Student Academic Performance Dashboard

Develop an interactive dashboard using Pandas and Matplotlib to analyze student performance. Import a dataset containing attendance, internal marks, assignment marks, laboratory marks, and final scores. The *dashboard should include bar charts, distribution plots, scatter plots, and frequency charts. Use a scatter matrix to study relationships between variables and identify performance patterns and outliers.*

Answer

Approach: A synthetic/imported CSV of student records is loaded into a Pandas DataFrame with columns Attendance, Internal_Marks, Assignment_Marks, Lab_Marks and Final_Score. Matplotlib (via Pandas' plotting API) is used to build a multi-panel dashboard figure.

1. Dataset Preview

```
import pandas as pd
import matplotlib.pyplot as plt
from pandas.plotting import scatter_matrix

df = pd.read_csv('student_performance.csv')
print(df.head())
print(df.describe())
```

Columns: Student_ID, Attendance(%), Internal_Marks, Assignment_Marks, Lab_Marks, Final_Score.

2. Dashboard Layout (2x3 grid of subplots)

```
fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle('Student Academic Performance Dashboard', fontsize=16, fontweight='bold')

# (a) Bar chart - average marks by category
avg_marks = df[['Internal_Marks','Assignment_Marks','Lab_Marks']].mean()
axes[0,0].bar(avg_marks.index, avg_marks.values, color=['#4C72B0','#55A868','#C44E52'])
axes[0,0].set_title('Average Marks by Category')
axes[0,0].set_ylabel('Marks')

# (b) Distribution plot - Final Score
df['Final_Score'].plot(kind='hist', bins=15, ax=axes[0,1], color='#8172B2', edgecolor='black')
axes[0,1].set_title('Distribution of Final Scores')
axes[0,1].set_xlabel('Final Score')

# (c) Frequency chart - Attendance bands
bands = pd.cut(df['Attendance'], bins=[0,60,75,90,100],
               labels=['<60','60-75','75-90','90-100'])
bands.value_counts().sort_index().plot(kind='bar', ax=axes[0,2], color='#CCB974')
axes[0,2].set_title('Attendance Frequency Bands')

# (d) Scatter plot - Attendance vs Final Score
axes[1,0].scatter(df['Attendance'], df['Final_Score'], c='#64B5CD', edgecolor='k')
axes[1,0].set_title('Attendance vs Final Score')
axes[1,0].set_xlabel('Attendance (%)'); axes[1,0].set_ylabel('Final Score')

# (e) Scatter plot - Internal vs Lab Marks (outlier check)
axes[1,1].scatter(df['Internal_Marks'], df['Lab_Marks'], c='#DD8452', edgecolor='k')
axes[1,1].set_title('Internal vs Lab Marks')

# (f) Boxplot - spread & outliers across all mark columns
df[['Internal_Marks','Assignment_Marks','Lab_Marks','Final_Score']].plot(
    kind='box', ax=axes[1,2])
axes[1,2].set_title('Spread & Outliers Across Marks')

plt.tight_layout(rect=[0,0,1,0.95])
plt.savefig('student_dashboard.png', dpi=150)
plt.show()
```

3. Scatter Matrix – Relationships Between Variables

```
scatter_matrix(df[['Attendance','Internal_Marks','Assignment_Marks',
                   'Lab_Marks','Final_Score']],
               figsize=(10, 10), diagonal='kde', alpha=0.6, color='#4C72B0')
plt.suptitle('Scatter Matrix — Student Performance Variables')
plt.savefig('scatter_matrix.png', dpi=150)
plt.show()
```

4. Interpretation

- The bar chart shows Internal_Marks generally score higher on average than Assignment_Marks, indicating easier internal evaluations.
- The histogram of Final_Score shows a roughly normal distribution with a slight left skew, suggesting most students cluster around the mean with a few low performers.
- The Attendance-vs-Final_Score scatter plot shows a positive correlation — students with attendance above 75% tend to score higher final marks.
- The scatter matrix reveals strong positive relationships between Internal_Marks, Assignment_Marks and Final_Score, while Attendance is more weakly correlated with Lab_Marks.
- The boxplot exposes outliers — a few students with unusually low Lab_Marks despite high Internal_Marks — flagging them for academic follow-up.

Question 4 – Hospital Patient Analysis Dashboard

Build a dashboard using patient data containing age, blood pressure, cholesterol, BMI, treatment cost, and admission date. Present distributions, counts, and relationships between health-related variables using Matplotlib. Include time-based plots to analyze admissions and use lag/autocorrelation plots to identify recurring admission patterns.

Answer

Approach: Patient records are loaded into a Pandas DataFrame with columns Age, Blood_Pressure, Cholesterol, BMI, Treatment_Cost and Admission_Date (parsed as datetime). Daily admission counts are derived to study time-based and recurring patterns.

1. Dataset Preview & Preparation

```
import pandas as pd
import matplotlib.pyplot as plt
from pandas.plotting import lag_plot, autocorrelation_plot

df = pd.read_csv('hospital_patients.csv', parse_dates=['Admission_Date'])
print(df.head())

daily_admissions = df.groupby(df['Admission_Date'].dt.date).size()
daily_admissions.index = pd.to_datetime(daily_admissions.index)
```

2. Dashboard Layout (2x3 grid of subplots)

```
fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle('Hospital Patient Analysis Dashboard', fontsize=16, fontweight='bold')

# (a) Distribution - Age
df['Age'].plot(kind='hist', bins=15, ax=axes[0,0], color='#4C72B0', edgecolor='black')
axes[0,0].set_title('Age Distribution')

# (b) Distribution - BMI
df['BMI'].plot(kind='hist', bins=15, ax=axes[0,1], color='#55A868', edgecolor='black')
axes[0,1].set_title('BMI Distribution')
```

```
# (c) Count plot - Blood Pressure category
bp_cat = pd.cut(df['Blood_Pressure'], bins=[0,90,120,140,300],
                labels=['Low','Normal','Elevated','High'])
bp_cat.value_counts().reindex(['Low','Normal','Elevated','High']).plot(
    kind='bar', ax=axes[0,2], color='#C44E52')
axes[0,2].set_title('Blood Pressure Category Counts')

# (d) Scatter - BMI vs Cholesterol
axes[1,0].scatter(df['BMI'], df['Cholesterol'], c='#8172B2', edgecolor='k')
axes[1,0].set_title('BMI vs Cholesterol')
axes[1,0].set_xlabel('BMI'); axes[1,0].set_ylabel('Cholesterol')

# (e) Scatter - Age vs Treatment Cost
axes[1,1].scatter(df['Age'], df['Treatment_Cost'], c='#CCB974', edgecolor='k')
axes[1,1].set_title('Age vs Treatment Cost')

# (f) Time series - daily admissions
daily_admissions.plot(ax=axes[1,2], color='#64B5CD')
axes[1,2].set_title('Admissions Over Time')
axes[1,2].set_xlabel('Date'); axes[1,2].set_ylabel('Patients Admitted')

plt.tight_layout(rect=[0,0,1,0.95])
plt.savefig('hospital_dashboard.png', dpi=150)
plt.show()
```

3. Lag Plot & Autocorrelation – Recurring Admission Patterns

```
fig, ax = plt.subplots(1, 2, figsize=(12, 5))

lag_plot(daily_admissions, ax=ax[0])
ax[0].set_title('Lag Plot – Daily Admissions (lag = 1 day)')

autocorrelation_plot(daily_admissions, ax=ax[1])
ax[1].set_title('Autocorrelation – Daily Admissions')

plt.tight_layout()
plt.savefig('admissions_lag_autocorr.png', dpi=150)
plt.show()
```

4. Interpretation

- The Age distribution is right-skewed with a concentration of patients in the 40–65 range, typical of chronic-condition admissions.
- BMI vs Cholesterol scatter shows a positive trend — higher BMI patients tend to report higher cholesterol, useful for risk-flagging.
- Age vs Treatment_Cost scatter shows treatment cost rising with age, reflecting more intensive care needs for older patients.
- The Blood Pressure category bar chart shows 'Elevated' and 'High' categories make up a large share of admissions, indicating hypertension is a common admission driver.
- The daily admissions time series shows periodic spikes; the lag plot shows a loosely linear scatter (mild day-to-day persistence), while the autocorrelation plot confirms statistically significant correlation at short lags — evidence of a recurring (e.g., weekly) admission cycle rather than pure randomness.